a Docking pipeline and the preparation choices it exposes
Input
PoseBusters set
428 complexes
Crystal pose
Receptor
pH 7.4
Cofactors removed
No waters in set
Held fixed
Ligand
Protonation
(supplied or pH 7.4)
Start conformer
Search space
Box padding
2 / 4 / 6 / 8 Å
Engine
AutoDock Vina
Smina
Gnina
rDock
Evaluation
RMSD < 2 Å
PoseBusters
validity
Held fixed
Varied one at a time
Engine
b One factor at a time, paired by complex
Vina
Smina
Gnina
rDock
Baseline (4 Å box)
Box padding 2 Å
Box padding 6 Å
Box padding 8 Å
Ligand protonation
Start conformer
one run: 428 complexes
Base
Arm
success
miss
outcome changed
c Which matters more: preparation or engine?
Engines, at a fixed preparation
Δ engine
Preparation choices, at a fixed engine
Δ preparation
Docking success rate